

Inference on Union Bounds

with Applications to DiD, RDD, Bunching, and Structural Counterfactuals

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January 11, 2024

Motivation

In many empirical applications, the target object is in a **union bound**

$$\theta \in \bigcup_{\beta \in \mathcal{B}} [\lambda_\ell(\beta), \lambda_u(\beta)]$$

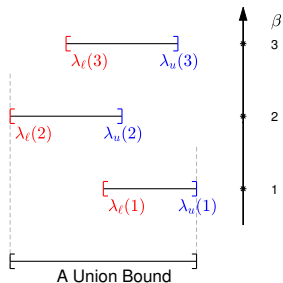
- θ is the target object
- $[\lambda_\ell(\beta), \lambda_u(\beta)]$ is a bound indexed by β
- $\mathcal{B}$ is the set of indices

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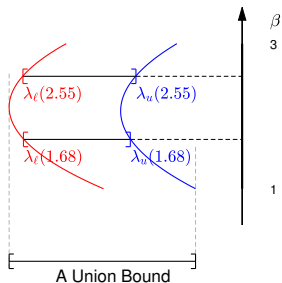


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The goal of this paper: Construct a *good* confidence interval for θ

Example with Finite $\mathcal{B}$: Sensitivity Analysis

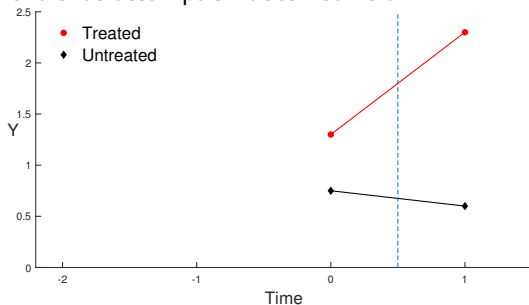
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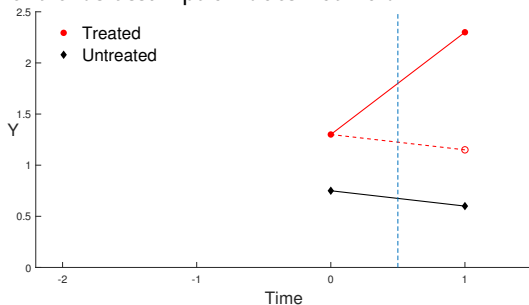


Units with $D = 1$ receive a treatment at $t = 1$

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$Y_t(d)$: the potential outcome at t with treatment d

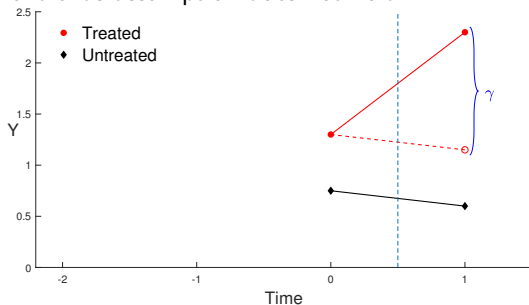
With parallel trends

$$0 = \mathbb{E} [Y_t(0) - Y_{t-1}(0) \mid D = 1] - \mathbb{E} [Y_t(0) - Y_{t-1}(0) \mid D = 0]$$

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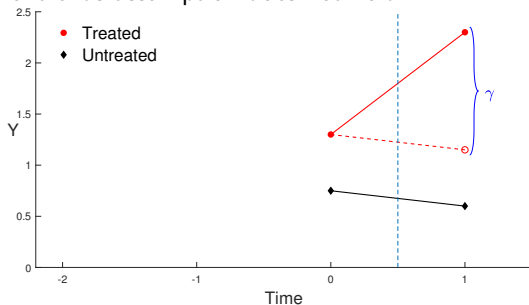
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$$\gamma = \underbrace{ATT}_{\theta}$$

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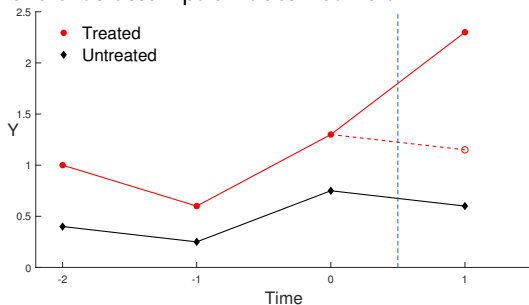
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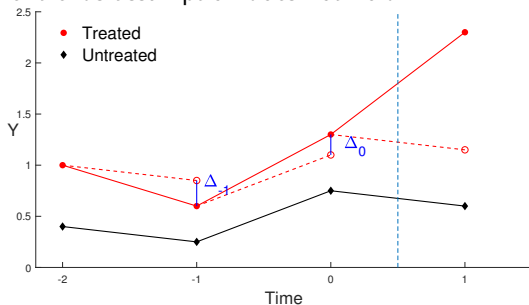
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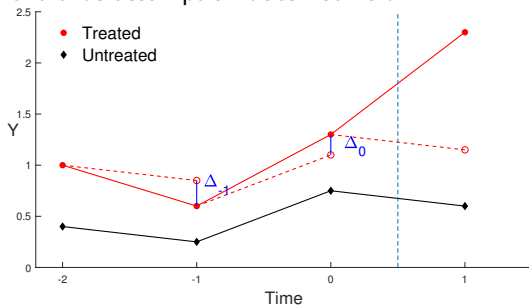
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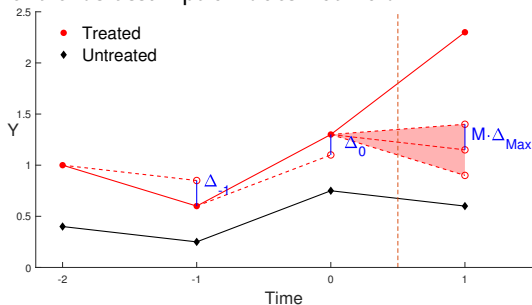
Relax the parallel trends assumption by

$$|\Delta_1| \leq M \cdot \max_{t=-T+1, \dots, 0} |\Delta_t|$$

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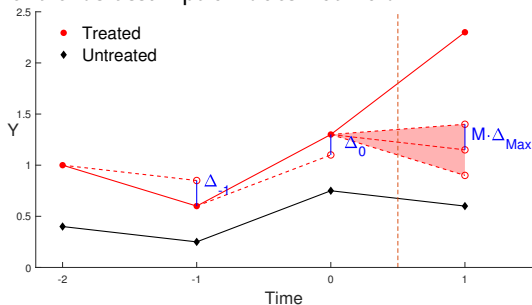
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$$\theta = ATT \in \bigcup_{\beta \in \mathcal{B}} [\lambda_\ell(\beta), \lambda_u(\beta)]$$

where $\mathcal{B} = \{-(T-1), \dots, T-1, T\}$,

$$[\lambda_\ell(\beta), \lambda_u(\beta)] = \begin{cases} [\gamma - M\Delta_\beta, \gamma + M\Delta_\beta] & \text{if } \beta = -(T-1), \dots, 0, \\ [\gamma + M\Delta_{\beta-T}, \gamma - M\Delta_{\beta-T}] & \text{if } \beta = 1, \dots, T \end{cases}$$

Example with Infinite $\mathcal{B}$: Counterfactual Analysis

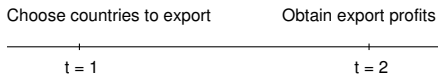
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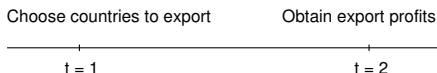
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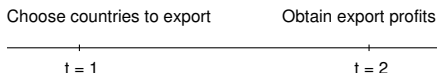
- β : structural parameter

$$\text{ExpProfit}(\beta)_{im} = \beta_1 \text{HomeRevenue}_i - \beta_2 - \beta_3 \text{Distance}_{im} - \varepsilon_{im}$$

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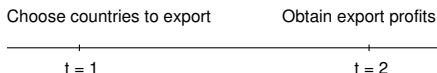
- $\mathcal{B}$: the set of β satisfying the revealed preference restriction

$$E[1[\text{export}] \times \text{ExpProfit}(\beta)] \geq 0$$

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$$E[1[\text{export}] \times \text{ExpProfit}(\beta)] \geq 0$$

- θ : the change in #exporters if info set, fixed cost changes
- $[\lambda_\ell(\beta), \lambda_u(\beta)]$: the bound of the change given potential true value β

(Some) Other Examples

Finite $\mathcal{B}$

- Difference-in-Differences: Hasegawa, Small, Webster (2019, Epidemiology), Ye, Keele, Hasegawa, and Small (2023, JASA), Ban and Kédagni (2023, WP)
- Regression Discontinuity Design: Kolesár and Rothe (2018, AER)
- Bunching and Income Elasticity: Blomquist, Newey, Kumar, and Liang (2021, JPE)
- Falsification Adaptive Set: Masten and Poirier (2021, ECTA), Stoye (2022, WP), Apfel and Windmeijer (2022, WP)
- Instrumental Variables: Machado, Shaikh, and Vytlačil (2019, JoE)

Infinite $\mathcal{B}$

- IO: Jia (2008, ECTA), Crawford and Yurukoglu (2012, AER), Eizenberg (2014, RES), Wollmann (2018, AER), Björkegren (2019, RES), Ciliberto, Murry, and Tamer (2021, JPE)
- Trade: Helpman, Itskhoki, Muendler, and Redding (2017, RES)
- Dynamic Disc. Choice: Kalouptsi, Kitamura, Lima, Souza-Rodrigues (2021, WP)
- Political: Kawai and Watanabe (2013, AER), Iaryczower, Shi, and Shum (2018, JPE), Bombardini, Li, and Trebbi (2023, AER)
- Treatment Effect: Mogstad, Santos, Torgovitsky (2018, ECTA), Ura (2018, QE)

Outline

- 1 Preview of the Contributions
- 2 Inference with Finite $\mathcal{B}$ (e.g. Difference in Differences)
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In this paper

- $\lambda_\ell(\beta), \lambda_u(\beta)$ are unknown but consistently estimable
- $\mathcal{B}$ is known or consistently estimable
- $\mathcal{B}$ can be *finite* or *infinite*

Finite $\mathcal{B}$: Main Contributions

I propose a novel CI based on modified conditional inference

- **Valid:** CI covers θ with prob. $\geq 1 - \alpha$ under mild regularity conditions
- **Short:** higher power than existing methods under a large set of DGPs

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Finite $\mathcal{B}$: Compare with Existing Procedures

My method improve upon existing valid procedures

	My	Simple	Hybrid	Adj. Boot.
Adjust for union	✓	×	×	✓
Apply to general cases	✓	✓	×	✓
$\sqrt{n}$ conv. rate to id set	✓	✓	✓	×

- Simple CI: Kolesár and Rothe (2018, AER), among others
- Hybrid CI: Rambachan and Roth (2023, RES)
- Adjusted Bootstrap: Ye, Keele, Hasegawa and Small (2023, JASA)

Finite $\mathcal{B}$: Contributions to Other Related Literature

Intersection Bounds & Moment Inequalities

Chernozhukov, Hong, and Tamer (2007), Romano and Shaikh (2008), Rosen (2008), D. Andrews and Guggenberger (2009), D. Andrews and Soares (2010), Chernozhukov, Lee, and Rosen (2013), D. Andrews and Shi (2013), Bugni, Canay and Shi (2015), among others

- This paper complements the intersection bounds

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Conditional Inference

I. Andrews and Mikusheva (2016), I. Andrews, Roth and Pakes (2016), I. Andrews, Kitagawa, McCloskey (2021, 2023), Rambachan and Roth (2023), among others

- This paper widens the use of the conditional inference technique

Finite $\mathcal{B}$: Empirical Illustration

RR23 in Dustmann, Lindner, Schönberg, Umkehrer, Vom Berge (2022, QJE)

What are the effects of the minimum wage?

- Addresses wage inequality
- Potential disemployment

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- Potential disemployment $\Leftarrow$ Insignificant Employment Effect

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- Is employment effect ≥ -0.6 (wage effect) w/o parallel trends? ► DiD

The authors relax parallel trends as in RR23 ► SDRM

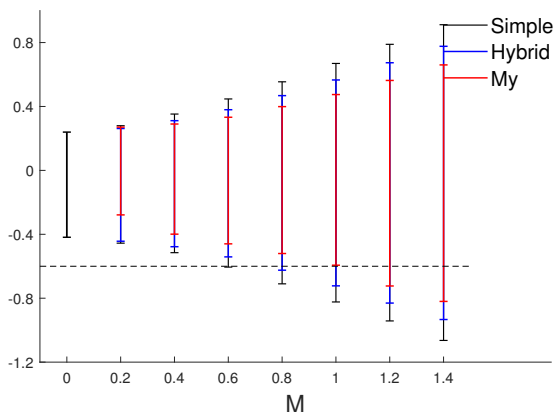
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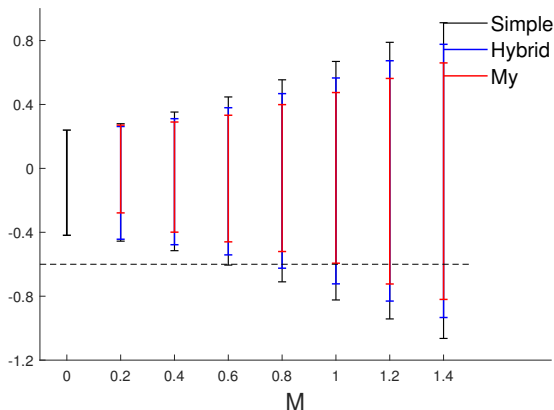
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$\mathcal{B}$ is formed by moment inequalities, e.g. structural counterfactuals

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Current practice

- Construct CS for structural parameters then project on the target object
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- × Not valid: no adjustment for the estimation uncertainty in $\hat{\lambda}_\ell(\beta), \hat{\lambda}_u(\beta)$

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Subvector Inference in Moment Inequalities

Bugni, Canay, Shi (2017), Chen, Chernozhukov, Tamer (2018), Kaido, Molinari, Stoye (2019), Cox and Shi (2023), I. Andrews, Roth, Pakes (2023), Chernozhukov, Newey, Santos (2023), Bei (2023)

- Subvector β_k does not have direct information for decision
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Counterfactual Analysis & Marginal Treatment Effect

Kalouptzidi, Kitamura, Lima, Souza-Rodrigues (2021), Cho and Russle (2023), Mogstad, Santos, Torgovitsky (2018)

- Depends on specific structures, e.g. dynamic discrete choice, linearity
- This paper: general moment inequality settings

Infinite $\mathcal{B}$: Empirical Illustration

Dickstein and Morales (2018, QJE) [► Details](#)

Question: What do exporters know?

One target object: If we change the information set from small to large, what is the percentage change in number of exporters?

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My	✓	✓	$[-20.4, -1.6]$	$[-27.4, -8.7]$	$[0, 1.2]$

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In this presentation, I will focus on the interval union bounds for finite $\mathcal{B}$

- Empirically motivated
- Same idea applies to non-connected union bounds (see the paper)

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Assume that the estimators $\hat{\lambda}_\ell, \hat{\lambda}_u$ are asymptotically normal

$$\sqrt{n} \begin{pmatrix} \hat{\lambda}_\ell - \lambda_\ell \\ \hat{\lambda}_u - \lambda_u \end{pmatrix} \xrightarrow{d} N(0, \Sigma)$$

Setting

The target object

$$\theta \in \left[\min_{\beta \in \mathcal{B}} \lambda_\ell(\beta), \max_{\beta \in \mathcal{B}} \lambda_u(\beta) \right]$$

where λ_ℓ and λ_u are $|\mathcal{B}|$ -dimensional vectors

Assume that the estimators $\hat{\lambda}_\ell, \hat{\lambda}_u$ are asymptotically normal

$$\sqrt{n} \begin{pmatrix} \hat{\lambda}_\ell - \lambda_\ell \\ \hat{\lambda}_u - \lambda_u \end{pmatrix} \xrightarrow{d} N(0, \Sigma)$$

Goal: construct a *uniformly valid* and *short* CI for θ

$$\liminf_n \inf_{P \in \mathcal{P}} \inf_{\theta \in [\lambda_{\ell, \min}, \lambda_{u, \max}]} P(\theta \in CI) \geq 1 - \alpha$$

A Simple Example

Consider the simplest possible case where

$$\theta \in [\min \{\lambda_1, \lambda_2\}, \max \{\lambda_1, \lambda_2\}]$$

with

$$\begin{pmatrix} \hat{\lambda}_1 \\ \hat{\lambda}_2 \end{pmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)$$

A Simple Example

Construct the CI by inverting tests of the hypothesis

$$H_0 : \min \{ \lambda_1, \lambda_2 \} \leq \theta \leq \max \{ \lambda_1, \lambda_2 \}$$

A Simple Example

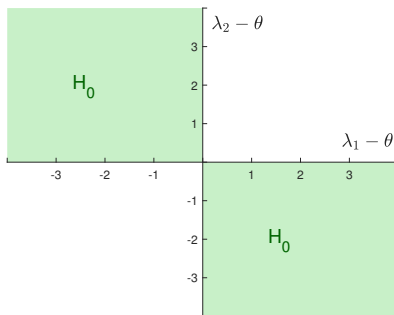
Construct the CI by inverting tests of the hypothesis

$$H_0 : \min \{ \lambda_1 - \theta, \lambda_2 - \theta \} \leq 0 \leq \max \{ \lambda_1 - \theta, \lambda_2 - \theta \}$$

A Simple Example

Construct the CI by inverting tests of the hypothesis

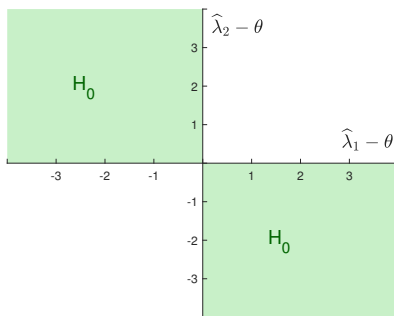
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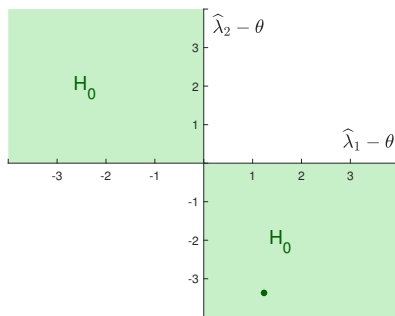
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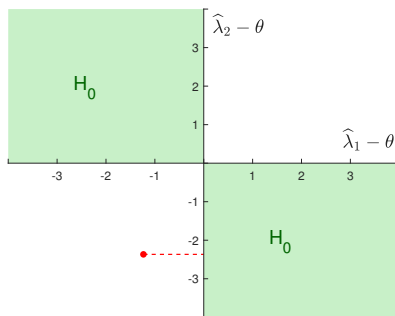
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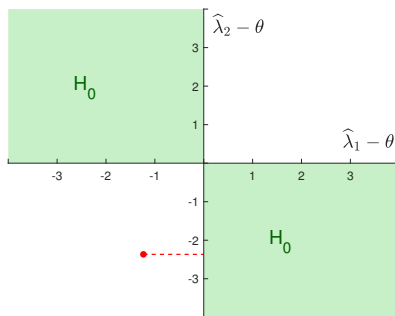
Test stat is

$$\hat{T}(\theta) = \max \left\{ \min \{ \hat{\lambda}_1 - \theta, \hat{\lambda}_2 - \theta \}, \min \{ \theta - \hat{\lambda}_1, \theta - \hat{\lambda}_2 \} \right\}$$

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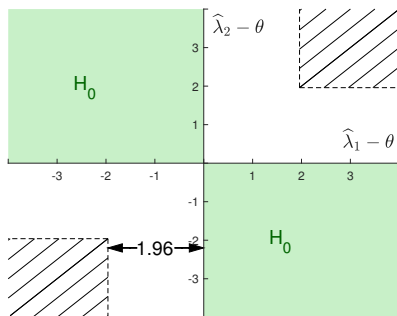
The CI is

$$\left[\min_{\hat{T}(\theta) \leq c(\theta)} \theta, \max_{\hat{T}(\theta) \leq c(\theta)} \theta \right]$$

A Simple Example

Construct the CI by inverting tests of the hypothesis

$$H_0 : \min \{ \lambda_1 - \theta, \lambda_2 - \theta \} \leq 0 \leq \max \{ \lambda_1 - \theta, \lambda_2 - \theta \}$$



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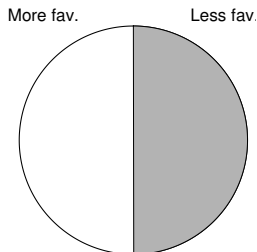
Current practice: $c^{\text{sim}} = \Phi^{-1}(1 - \frac{\alpha}{2})$, 1.96 for $\alpha = 0.05$

A Simple Example - Modified Conditional CV

To improve upon c^{sim} , I propose a modified conditional cv

A Simple Example - Modified Conditional CV

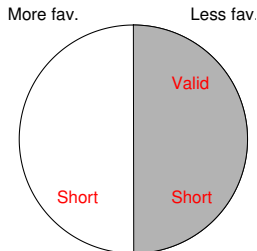
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1. Define less and more favorable DGPs

A Simple Example - Modified Conditional CV

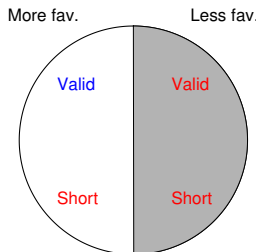
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1. Define less and more favorable DGPs
2. Construct a **conditional** cv

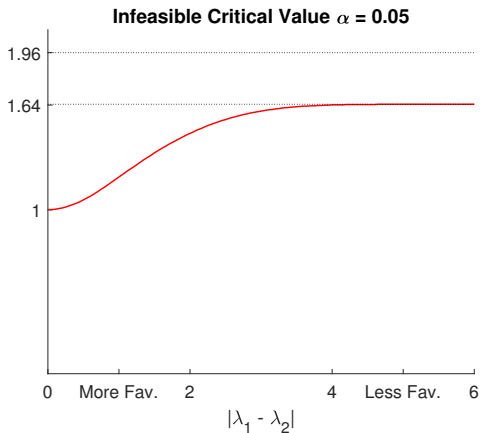
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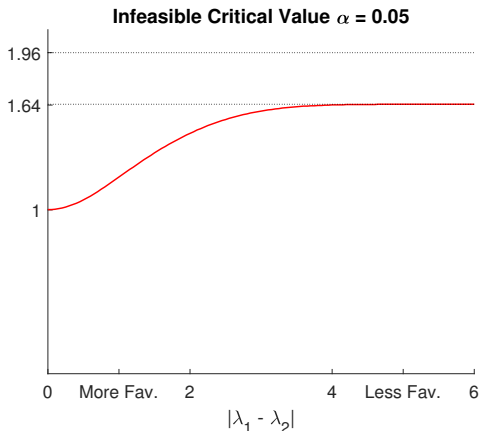


1. Define less and more favorable DGPs
2. Construct a **conditional** cv
3. **Modify** the conditional cv

A Simple Example - Step 1 More & Less fav. DGPs

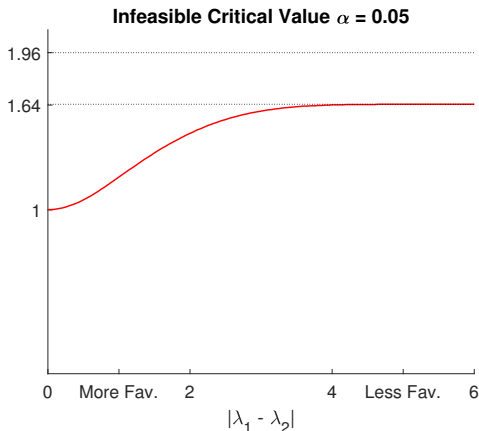


A Simple Example - Step 1 More & Less fav. DGPs



If $\lambda_1 = \lambda_2$, the infeasible CI is $\left[\min\{\hat{\lambda}_1, \hat{\lambda}_2\} - 1, \max\{\hat{\lambda}_1, \hat{\lambda}_2\} + 1 \right]$

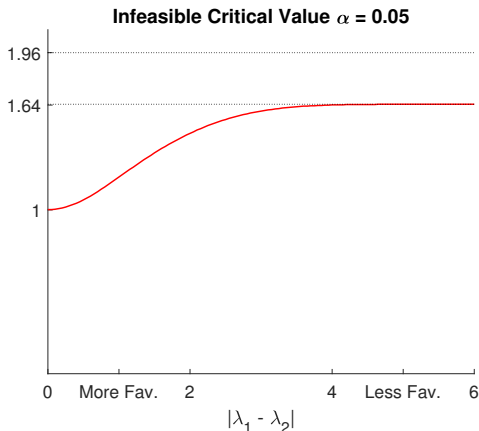
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- we can use $\left[\hat{\lambda}_1 - 1.96, \hat{\lambda}_1 + 1.96 \right]$ or $\left[\hat{\lambda}_2 - 1.96, \hat{\lambda}_2 + 1.96 \right]$

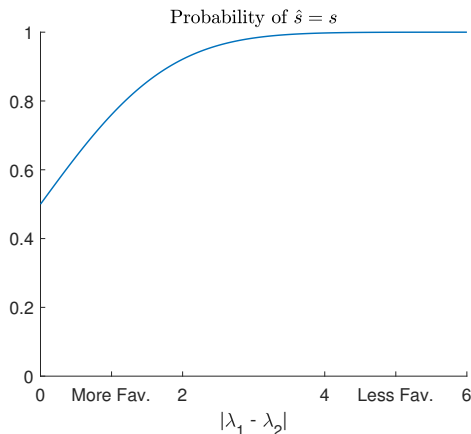
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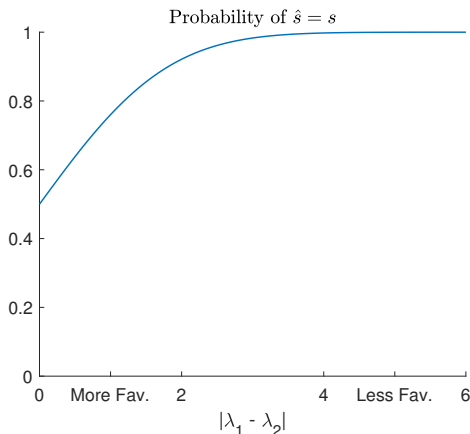
If $\lambda_1 \ll \lambda_2$, the infeasible CI is $\left[\min\{\hat{\lambda}_1, \hat{\lambda}_2\} - 1.64, \max\{\hat{\lambda}_1, \hat{\lambda}_2\} + 1.64 \right]$

A Simple Example - Step 2 Conditional CV



where $s = \arg \min_{\beta=1,2} \lambda_{\beta}$, $\hat{s} = \arg \min_{\beta=1,2} \hat{\lambda}_{\beta}$

A Simple Example - Step 2 Conditional CV



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Construct conditional cv based on the distribution of $\hat{T}(\theta) \big| \hat{s} = s$

A Simple Example - Step 2 Conditional CV

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Recall

$$\hat{T}(\theta) = \max \left\{ \min \left\{ \theta - \hat{\lambda}_1, \theta - \hat{\lambda}_2 \right\}, \min \left\{ \hat{\lambda}_1 - \theta, \hat{\lambda}_2 - \theta \right\} \right\}$$

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Specifically, I consider

$$\hat{T}(\theta) \big| \hat{T}(\theta) = \theta - \hat{\lambda}_1, \hat{s} = s$$

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A Simple Example - Step 2 Conditional CV

Construct conditional cv based on the distribution of $\hat{T}(\theta) \mid \hat{s} = s$

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A Simple Example - Step 2 Conditional CV

I illustrate with

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$$\hat{T}(\theta) \mid \hat{T}(\theta) = \hat{\lambda}_1 - \theta, \hat{s} = s \quad \overset{\text{FOSD}}{\preceq} \quad \mathcal{TN}\left(0, \left[\theta - \hat{\lambda}_2, \hat{\lambda}_2 - \theta\right]\right)$$

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$$\widehat{T}(\theta) \Big| \widehat{T}(\theta) = \widehat{\lambda}_1 - \theta, \widehat{s} = s \quad \overset{\text{FOSD}}{\preceq} \quad \mathcal{TN}\left(0, \left[\theta - \widehat{\lambda}_2, \widehat{\lambda}_2 - \theta\right]\right)$$

Let $\alpha^{\text{con}} \in (\frac{\alpha}{2}, \alpha)$ be the conditional rejection rate, recommend $\alpha^{\text{con}} = \frac{4}{5}\alpha$

Define $\widehat{c}^{\text{con}}$ as the $1 - \alpha^{\text{con}}$ quantile of $\mathcal{TN}\left(0, \left[\theta - \widehat{\lambda}_2, \widehat{\lambda}_2 - \theta\right]\right)$

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✓ Valid size under less favorable DGPs:

$$P\left(\widehat{T}(\theta) > \widehat{c}^{\text{con}}\right) = P\left(\widehat{T}(\theta) > \widehat{c}^{\text{con}}, \widehat{s} = s\right) + P\left(\widehat{T}(\theta) > \widehat{c}^{\text{con}}, \widehat{s} \neq s\right)$$

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✓ Smaller than c^{sim}

$$\hat{c}^{\text{con}} = \Phi^{-1}\left(\alpha^{\text{con}} + (1 - 2\alpha^{\text{con}})\Phi\left(\hat{\lambda}_2 - \theta\right)\right)$$

A Simple Example - Step 2 Conditional CV

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$$\hat{T}(\theta) \mid \hat{T}(\theta) = \hat{\lambda}_1 - \theta, \hat{s} = s \quad \overset{\text{FOSD}}{\preceq} \quad \mathcal{TN}\left(0, [\theta - \hat{\lambda}_2, \hat{\lambda}_2 - \theta]\right)$$

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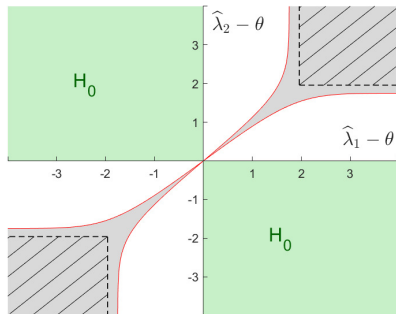
$$P\left(\hat{T}(\theta) > \hat{c}^{\text{con}}\right) \leq P\left(\hat{T}(\theta) > \hat{c}^{\text{con}} \mid \hat{s} = s\right) P(\hat{s} = s) + P(\hat{s} \neq s) \leq \alpha$$

✓ Smaller than c^{sim}

$$\begin{aligned} \hat{c}^{\text{con}} &= \Phi^{-1}\left(\alpha^{\text{con}} + (1 - 2\alpha^{\text{con}})\Phi\left(\hat{\lambda}_2 - \theta\right)\right) \\ &\leq \Phi^{-1}\left(\alpha^{\text{con}} + (1 - 2\alpha^{\text{con}})\right) \\ &= \Phi^{-1}\left(1 - \alpha^{\text{con}}\right) < \Phi^{-1}\left(1 - \frac{\alpha}{2}\right) = c^{\text{sim}} \end{aligned}$$

A Simple Example - Step 2 Conditional CV

The rejection region of $\hat{c}^{\text{con}}$ for $H_0 : \min \{\lambda_1, \lambda_2\} \leq \theta \leq \max \{\lambda_1, \lambda_2\}$



A Simple Example - Step 3 Modification

I introduce a novel modification

$$c^m(\theta, c^t) = \begin{cases} \hat{c}^{con}(\theta) & \text{if } \hat{c}^{con}(\theta) \geq c^t \\ c^t & \text{if } \hat{c}^{con}(\theta) < c^t \end{cases}$$

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To determine the value of c^t , fix $c^t = c$

$$P(\theta \notin CI; \lambda) \leq \max \{P([\lambda_s, \lambda_{mid}] \not\subseteq CI; \lambda), P([\lambda_{mid}, \lambda_b] \not\subseteq CI; \lambda)\}$$

with $\lambda = (\lambda_1, \lambda_2)$, $\lambda_s = \min\{\lambda_1, \lambda_2\}$, $\lambda_b = \max\{\lambda_1, \lambda_2\}$, $\lambda_{mid} = (\lambda_1 + \lambda_2)/2$

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$$P(\theta \notin CI; \lambda) \leq \max \{ P([\lambda_s, \lambda_{mid}] \not\subseteq CI; \lambda), P([\lambda_{mid}, \lambda_b] \not\subseteq CI; \lambda) \}$$

because $\theta \in [\lambda_s, \lambda_{mid}]$ or $\theta \in [\lambda_{mid}, \lambda_b]$

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To determine the value of c^t , fix $c^t = c$

$$\begin{aligned} P(\theta \notin CI; \lambda) &\leq \max \{ P([\lambda_s, \lambda_{mid}] \not\subseteq CI; \lambda), P([\lambda_{mid}, \lambda_b] \not\subseteq CI; \lambda) \} \\ &\leq \max \{ P(T(\lambda_s) > c^m(\lambda_s, c) \text{ or } T(\lambda_{mid}) > c^m(\lambda_{mid}, c); \lambda), \\ &\quad P(T(\lambda_b) > c^m(\lambda_b, c) \text{ or } T(\lambda_{mid}) > c^m(\lambda_{mid}, c); \lambda) \} \end{aligned}$$

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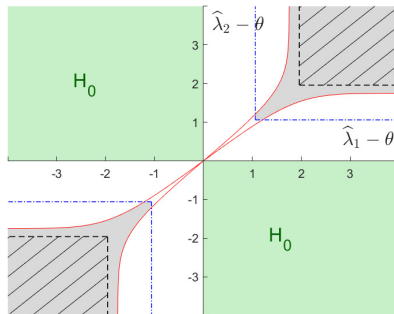
Thus it suffices to have

$$c^t = \inf \left\{ c \geq 0 : \sup_{\lambda \in \Lambda} \bar{p}(c, \lambda) \leq \alpha \right\}$$

This lower truncation guarantees uniform coverage

A Simple Example - Step 3 Modification

The rejection region of $\hat{c}^m$ for $H_0 : \min \{ \lambda_1, \lambda_2 \} \leq \theta \leq \max \{ \lambda_1, \lambda_2 \}$



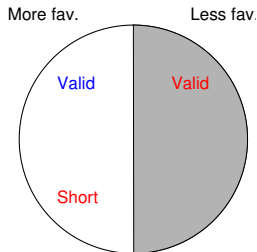
Larger power: the new CI has a larger rejection region

Valid: the lower truncation removes counter-intuitive rejection region

A Simple Example - Summary

The main idea of the modified conditional CI

1. Define less and more favorable DGPs
2. Construct a **conditional** cv
 - valid under less favorable DGPs
3. **Modify** the conditional cv
 - valid under more favorable DGPs



General Cases - Test Statistics

The target object

$$\theta \in \left[\min_{b \in \mathcal{B}} \lambda_{\ell, b}, \max_{b \in \mathcal{B}} \lambda_{u, b} \right]$$

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$$\sqrt{n} \begin{pmatrix} \hat{\lambda}_{\ell} - \lambda_{\ell} \\ \hat{\lambda}_u - \lambda_u \end{pmatrix} \xrightarrow{d} N(0, \Sigma), \quad \hat{\Sigma} \xrightarrow{p} \Sigma$$

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Construct CI by inverting

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Construct CI by inverting

$$H_0 : \min_{b \in \mathcal{B}} \lambda_{\ell, b} \leq \theta \leq \max_{b \in \mathcal{B}} \lambda_{u, b}$$

The test statistic

$$\hat{T}(\theta) = \max \left\{ \min_{b \in \mathcal{B}} \frac{\sqrt{n}(\hat{\lambda}_{\ell, b} - \theta)}{\hat{\sigma}_{\ell, b}}, \min_{b \in \mathcal{B}} \frac{\sqrt{n}(\theta - \hat{\lambda}_{u, b})}{\hat{\sigma}_{u, b}} \right\}$$

where $\hat{\sigma}_{\ell, b} = \sqrt{\text{var}(\hat{\lambda}_{\ell, b})}$, $\hat{\sigma}_{u, b} = \sqrt{\text{var}(\hat{\lambda}_{u, b})}$

General Cases - Conditional Critical Value

The conditional critical value

$$\hat{c}^{con} = \begin{cases} \Phi^{-1} \left(\alpha^{con} \Phi \left(t_{\ell,1}(\theta, \hat{b}_\ell) \right) + (1 - \alpha^{con}) \Phi \left(t_{\ell,2}(\theta, \hat{b}_\ell) \right) \right) & \text{if } \hat{T}(\theta) = \frac{\hat{\lambda}_{\ell, \hat{b}_\ell} - \theta}{\hat{\sigma}_{\ell, \hat{b}_\ell} / \sqrt{n}} \\ \Phi^{-1} \left(\alpha^{con} \Phi \left(t_{u,1}(\theta, \hat{b}_u) \right) + (1 - \alpha^{con}) \Phi \left(t_{u,2}(\theta, \hat{b}_u) \right) \right) & \text{if } \hat{T}(\theta) = \frac{\theta - \hat{\lambda}_{u, \hat{b}_u}}{\hat{\sigma}_{u, \hat{b}_u} / \sqrt{n}} \end{cases}$$

where

$$\hat{b}_\ell = \arg \min_{b \in \mathcal{B}} \frac{\hat{\lambda}_{\ell, b} - \theta}{\hat{\sigma}_{\ell, b} / \sqrt{n}}, \quad \hat{b}_u = \arg \min_{b \in \mathcal{B}} \frac{\theta - \hat{\lambda}_{u, b}}{\hat{\sigma}_{u, b} / \sqrt{n}}$$

$$t_{\ell,1}(\theta, b) = \min_{\tilde{b} \in \mathcal{B}} (1 + \hat{\rho}_{\ell u}(b, \tilde{b}))^{-1} \left(\frac{\theta - \hat{\lambda}_{u, b}}{\hat{\sigma}_{u, b} / \sqrt{n}} + \hat{\rho}_{\ell u}(b, \tilde{b}) \frac{\hat{\lambda}_{\ell, b} - \theta}{\hat{\sigma}_{\ell, b} / \sqrt{n}} \right)$$

$$t_{\ell,2}(\theta, b) = \min_{\tilde{b} \in \mathcal{B}} (1 - \hat{\rho}_\ell(b, \tilde{b}))^{-1} \left(\frac{\hat{\lambda}_{\ell, b} - \theta}{\hat{\sigma}_{\ell, b} / \sqrt{n}} - \hat{\rho}_\ell(b, \tilde{b}) \frac{\hat{\lambda}_{\ell, b} - \theta}{\hat{\sigma}_{\ell, b} / \sqrt{n}} \right)$$

$$\hat{\rho}_\ell(b_1, b_2) = \frac{\hat{\sigma}_{\ell, b_1 b_2}}{\hat{\sigma}_{\ell, b_1} \hat{\sigma}_{\ell, b_2}}, \quad \hat{\rho}_u(b_1, b_2) = \frac{\hat{\sigma}_{u, b_1 b_2}}{\hat{\sigma}_{u, b_1} \hat{\sigma}_{u, b_2}}, \quad \hat{\rho}_{\ell u}(b_1, b_2) = \frac{\hat{\sigma}_{\ell u, b_1 b_2}}{\hat{\sigma}_{\ell, b_1} \hat{\sigma}_{u, b_2}}$$

General Cases - Conditional Critical Value

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- If $\hat{T}(\theta) = \frac{\hat{\lambda}_{\ell, \hat{b}_{\ell}} - \theta}{\hat{\sigma}_{\ell, \hat{b}_{\ell}} / \sqrt{n}}$, conditional on $\arg \min_{b \in \mathcal{B}} \frac{\hat{\lambda}_{\ell, b} - \theta}{\hat{\sigma}_{\ell, b} / \sqrt{n}} = \arg \min_{b \in \mathcal{B}} \frac{\lambda_{\ell, b} - \theta}{\sigma_{\ell, b} / \sqrt{n}}$

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- If identified set large, $\max_{b \in \mathcal{B}} \lambda_{u, b} - \min_{b \in \mathcal{B}} \lambda_{\ell, b} \gg \frac{1}{\sqrt{n}}$,

$$\hat{c}^{con} \leq \Phi^{-1} (1 - \alpha^{con}) < c^{sim}$$

- In addition, if the bounds are not well separated, $\downarrow \hat{c}^{con}$

General Cases - Modified Conditional Critical Value

The modified conditional critical value

$$c^m(\theta, c^t) = \begin{cases} \hat{c}^{\text{con}}(\theta) & \text{if } \hat{c}^{\text{con}}(\theta) \geq c^t \\ c^t & \text{if } \hat{c}^{\text{con}}(\theta) < c^t \end{cases}$$

The lower truncation c^t is

$$c^t = \inf_c \left\{ c \geq 0 : \sup_{\lambda \in \Lambda} \bar{p}(c, \lambda) \leq \alpha \right\}$$

where $\theta_\ell = \min_{b \in \mathcal{B}} \lambda_{\ell, b}$, $\theta_u = \max_{b \in \mathcal{B}} \lambda_{u, b}$, $\theta_m = (\theta_\ell + \theta_u)/2$

$$\begin{aligned} \bar{p}(c, \lambda) = \max \bigg\{ & P \left(\hat{T}(\theta_\ell) > c^m(\theta_\ell, c) \text{ or } \hat{T}(\theta_m) > c^m(\theta_m, c); (\lambda, \hat{\Sigma}) \right), \\ & P \left(\hat{T}(\theta_m) > c^m(\theta_m, c) \text{ or } \hat{T}(\theta_u) > c^m(\theta_u, c); (\lambda, \hat{\Sigma}) \right) \bigg\} \end{aligned}$$

Size & Power: Asymptotic Size Properties

Assumption

- 1 *Asymptotic Normality (AN)*. $(\hat{\lambda}_\ell, \hat{\lambda}_u) \xrightarrow{d} \mathcal{N}(0, \Sigma)$
- 2 *Known Singularity (KS)*. $\Sigma = A\Omega A'$, where A is known and Ω has full rank
- 3 *Consistent Estimator (CE)*. Ω can be consistently estimated

Theorem (Uniform Coverage)

Suppose Assumptions AN, KS, CE hold, for any $\alpha \in (0, 0.5)$,

$$\liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}} \inf_{\theta \in [\lambda_{\ell, b_\ell}, \lambda_{u, b_u}]} P(\theta \in CI^m) \geq 1 - \alpha$$

The modified conditional CI has proper asymptotic coverage

► Assumptions

Size & Power: Comparison with Simple CI

Theorem (Symmetric Or Large Bounds)

Suppose Assumptions *KS*, *AN*, *FR*, *CE* hold, $\alpha \in (0, \frac{1}{2})$

(Symmetric Bounds) If $\text{corr}(\hat{\lambda}_{\ell, b_1}, \hat{\lambda}_{\ell, b_2}) < \rho_1^*(\alpha, \alpha^c)$, $\text{corr}(\hat{\lambda}_{\ell, b_\ell}, \hat{\lambda}_{\ell, b_u}) < \rho_2^*(\alpha)$

$$\hat{\lambda}_\ell = \hat{\lambda}_u$$

Then

❶ *My CI is Strictly Shorter:*

There is $\alpha' > \alpha$ such that

$$\liminf_n P \left(CI^m(\hat{\lambda}_n, \hat{\Sigma}_n/n; \alpha) \subseteq CI^{sim}(\hat{\lambda}_n, \hat{\Sigma}_n/n; \alpha') \right) = 1$$

❷ *My CI has Higher Power:*

There is $\kappa \in (0, +\infty)$ such that

$$\liminf_n P \left(\theta_n \notin CI^m(\hat{\lambda}, \hat{\Sigma}/n; \alpha) \right) - P \left(\theta_n \notin CI^{sim}(\hat{\lambda}, \hat{\Sigma}/n; \alpha) \right) > 0$$

for some $\theta_n = \theta_\ell - \frac{\kappa}{\sqrt{n}}$.

Size & Power: Comparison with Simple CI

Theorem (Symmetric Or Large Bounds)

Suppose Assumptions *KS*, *AN*, *FR*, *CE* hold, $\alpha \in (0, \frac{1}{2})$

(Large Bounds)

$$\max_{b \in \mathcal{B}} \lambda_{u,b} - \min_{b \in \mathcal{B}} \lambda_{\ell,b} \geq \varepsilon > 0$$

Then

❶ *My CI is Strictly Shorter:*

There is $\alpha' > \alpha$ such that

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for some $\theta_n = \theta_\ell - \frac{\kappa}{\sqrt{n}}$.

Size & Power: Comparison with Adj Boot CI

Let $CI^{AdjBoot}$ be the adjusted bootstrap procedure proposed in YKHS23

- use a subsample with size $m = \frac{n}{\kappa_n}$, $\kappa_n \rightarrow \infty$ and $\kappa_n = o(n)$
- use the full sample to estimate the nuisance para., e.g. $\sqrt{m}(\lambda_{\ell,b} - \min \lambda_\ell)$

Theorem (Larger Local Power)

Let $a > 0$, $\kappa'_n = o(\sqrt{\kappa_n})$, $\kappa'_n \rightarrow \infty$. Then

$$\liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n} \left\{ P(\theta_n \notin CI^m) - P(\theta_n \notin CI^{AdjBoot}) \right\} \geq 1 - \alpha$$

for local alternatives

$$\theta_n = \min_{b \in \mathcal{B}} \lambda_{\ell,b} - \frac{\kappa'_n}{\sqrt{n}} a \text{ or } \theta_n = \max_{b \in \mathcal{B}} \lambda_{u,b} + \frac{\kappa'_n}{\sqrt{n}} a$$

Simulation - Setting

Relaxation of the parallel trend assumptions, where

$$\theta = ATT \in \left[\min_{b \in \mathcal{B}} \lambda_{\ell, b}, \max_{b \in \mathcal{B}} \lambda_{u, b} \right]$$

where $\mathcal{B} = \{-(T-1), \dots, T\}$,

$$\lambda_{\ell} = \lambda_u = \begin{cases} \gamma + \Delta_{\beta} & \text{if } \beta = -(T-1), \dots, 0 \\ \gamma - \Delta_{-(\beta-1)} & \text{if } \beta = 1, \dots, T \end{cases}$$

Simulation - Setting

I conduct inference based on $(\hat{\Delta}, \hat{\gamma}, \Omega)$ where $(\hat{\Delta}, \hat{\gamma})$ is simulated from

$$\begin{pmatrix} \hat{\Delta} \\ \hat{\gamma} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \Delta \\ \gamma \end{pmatrix}, \Omega\right)$$

I use four Ω calibrated from

- 1 Benzarti and Carloni (2019): consumption tax cuts
- 2 Dustmann, Lindner, Schönberg, Umkehrer, Vom Berge (2022): minimum wage
- 3 Lovenheim and Willén (2019): teacher collective bargaining
- 4 Christensen, Keiser, Lade (2023): environmental crises

I normalized $\gamma = 0$ and use three Δ

- 1 Parallel trends assumption holds, i.e. $\Delta = 0_T$
- 2 Small pre-trends: Δ is calibrated
- 3 One large pre-trend: $\Delta = (10\omega_M, 0_{T-1})$

In sum, I use $4 \times 3 = 12$ empirically motivated DGPs [► Details](#)

Simulation - Alternative Methods

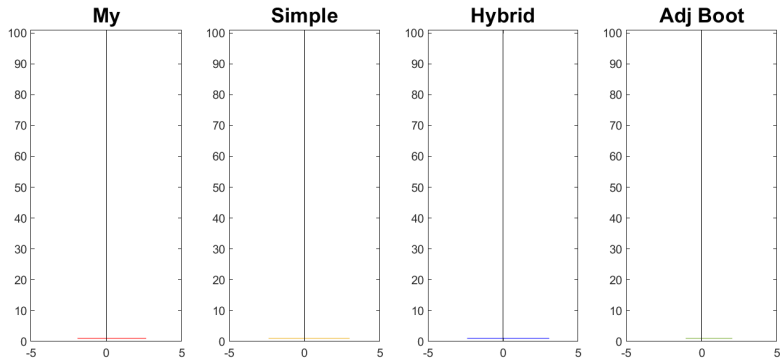
I compare my CI with

- ➊ Simple CI in, e.g., Kolesár and Rothe (2018, AER)
- ➋ Hybrid CI in Rambachan and Roth (2023, RES)
- ➌ Adjusted bootstrap in Ye, Keele, Hasegawa and Small (2023, JASA)

Simulation - Confidence Intervals

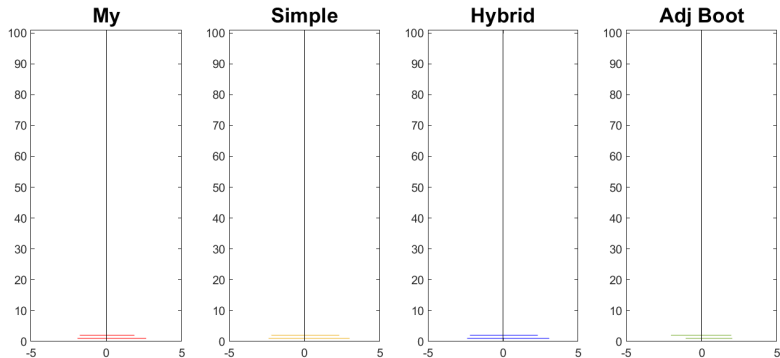
Simulation - Confidence Intervals

DGP: $\Delta = 0$, Ω from Lovenheim and Willén (2019)



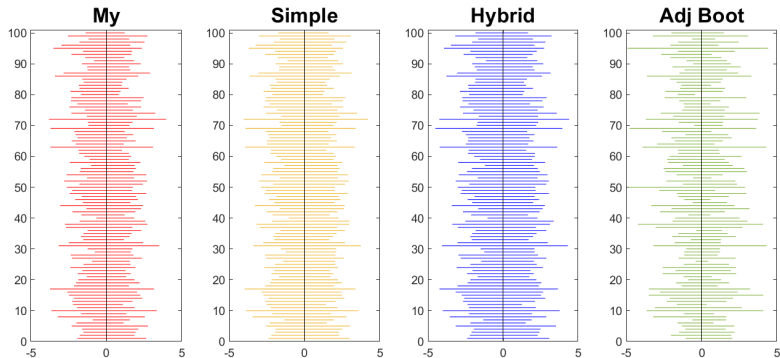
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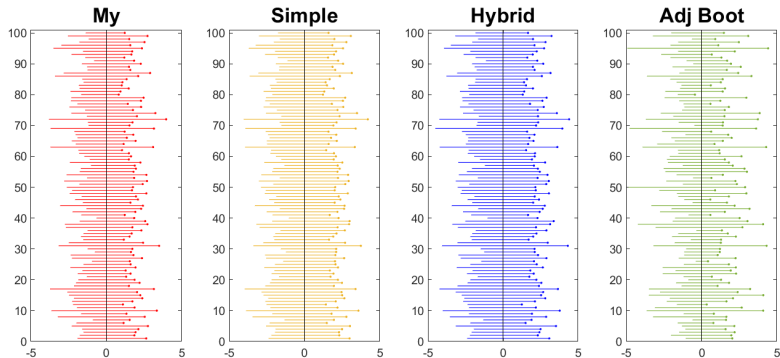
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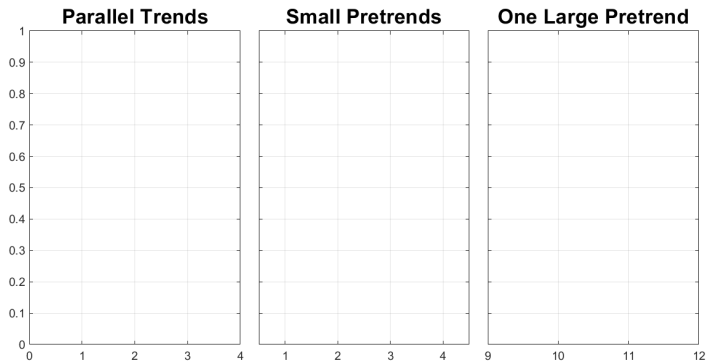
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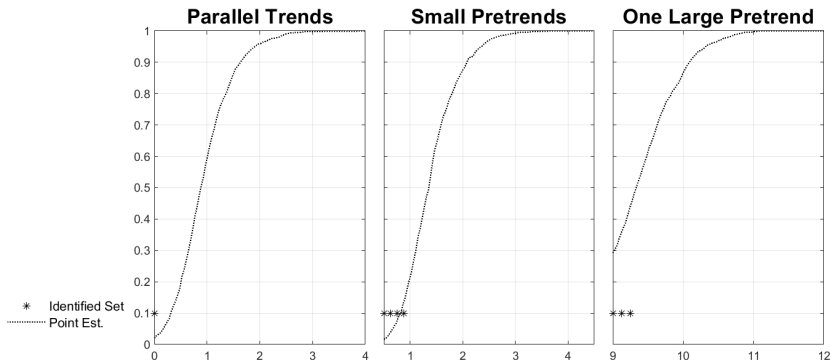
Simulation - CDF of Upper Bounds of CIs

DGP: Ω from Lovenheim and Willén (2019)



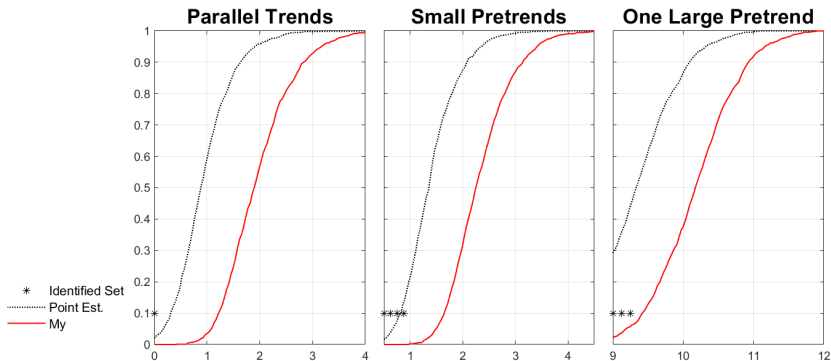
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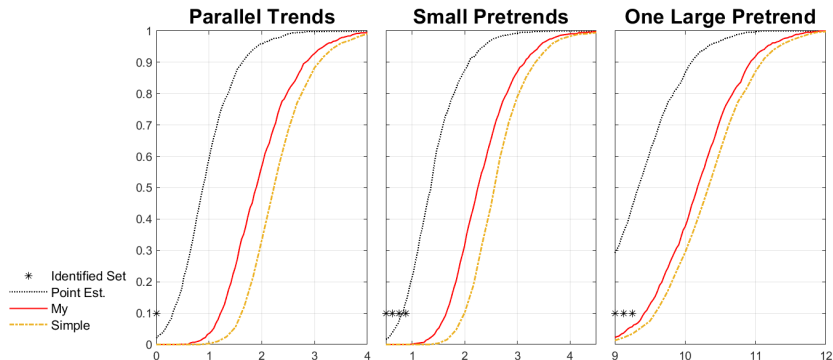
DGP: Ω from Lovenheim and Willén (2019)



Modified conditional CI has proper coverage

Simulation - CDF of Upper Bounds of CIs

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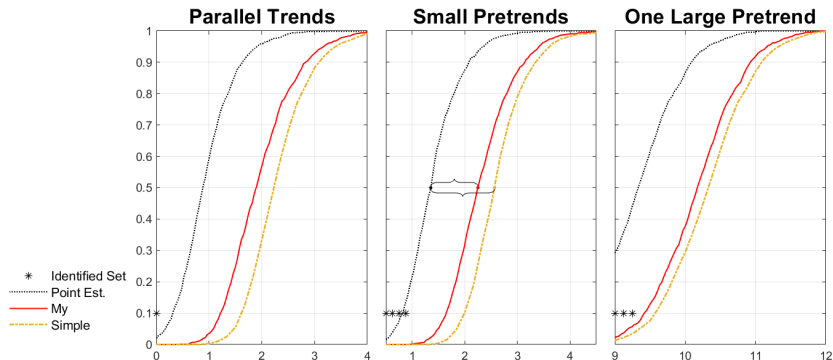


Modified conditional CI outperforms simple CI in all DGPs

- Reduces median simple CI (net of point est.) by **31%** under small violation

Simulation - CDF of Upper Bounds of CIs

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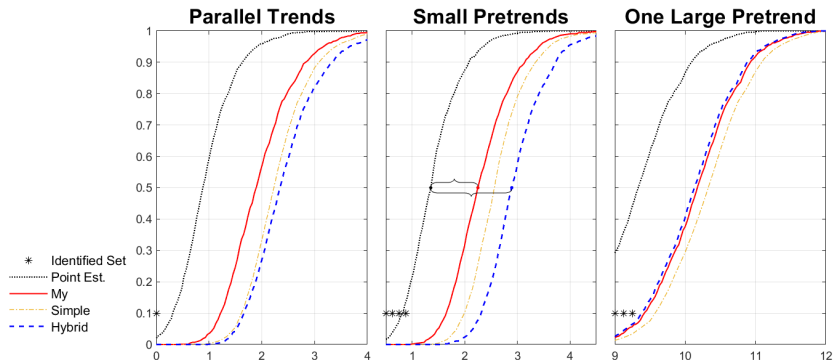


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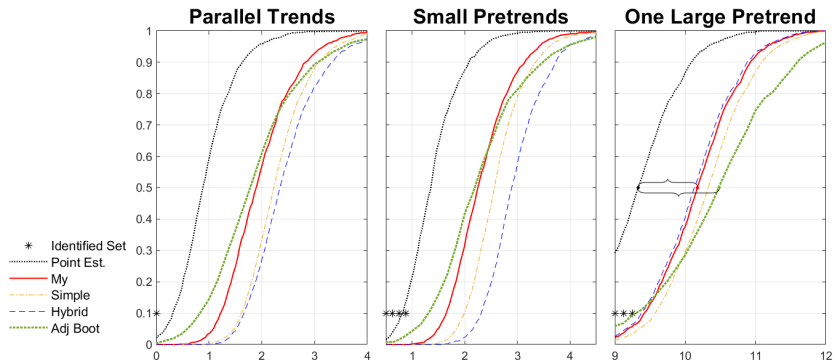


Modified conditional CI outperforms Hybrid CI under no or small violations

- Reduces median Hybrid CI (net of point est.) by **43%** under small violation
- Hybrid CI is efficient with large violation, but my CI is close

Simulation - CDF of Upper Bounds of CIs

DGP: Ω from Lovenheim and Willén (2019)

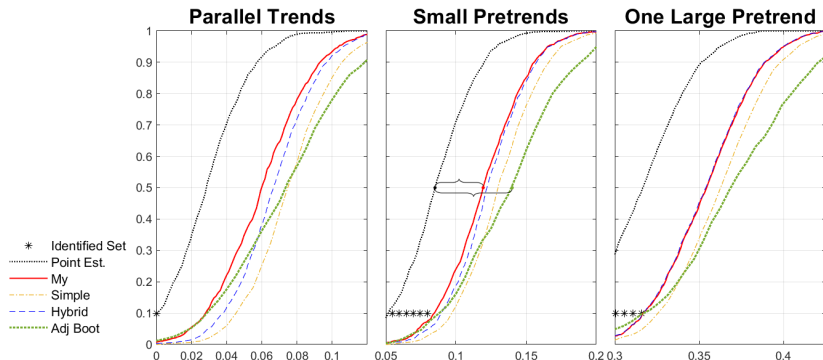


Modified conditional CI outperforms YKHS for relatively large alternatives

- Reduces median Adj Boot CI (net of point est.) by **27%** under large violation
- Power plot of Adj Boot will be flatter with larger n

Simulation - CDF of Upper Bounds of CIs

DGP: Ω from Benzarti and Carloni (2019)



Modified conditional CI outperforms Adj Boot CI for relatively large alternatives

- Reduces median Adj Boot CI (net of point est.) by **38%** under small violation
- Power plot of Adj Boot CI will be flatter with larger n

Outline

- 1 Preview of the Contributions
- 2 Inference with Finite $\mathcal{B}$ (e.g. Difference in Differences)
- 3 Inference with Infinite $\mathcal{B}$ (e.g. Structural Counterfactual)
- 4 Conclusion

Setting

The target object

$$\theta \in \bigcup_{\beta \in \mathcal{B}} [\lambda_\ell(\beta), \lambda_u(\beta)]$$

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$$E[m_j(W; \beta)] \leq 0, \quad j = 1, \dots, J \quad (\text{ineq})$$

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- $\mathcal{B}$ is the identified set of β

$$\mathcal{B} = \{\beta \in \overline{\mathcal{B}} : (\text{ineq}) \text{ holds}\}$$

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$$\mathcal{B} = \{\beta \in \overline{\mathcal{B}} : (\text{ineq}) \text{ holds}\}$$

- λ_ℓ and λ_u can be written as functions of moments

$$\lambda_\ell(\beta) = \lambda_\ell^\dagger(E[m_\lambda(W; \beta)])$$

$$\lambda_u(\beta) = \lambda_u^\dagger(E[m_\lambda(W; \beta)])$$

► Example

Calibrated Projection

Current practice

- Get $1 - \alpha$ confidence set of β

$$\tilde{\mathcal{B}} = \left\{ \beta \in \overline{\mathcal{B}} : \frac{\sqrt{n} \bar{m}_j(\beta)}{\hat{\sigma}_j(\beta)} \leq \tilde{c}(\beta), \quad \text{for all } j = 1, \dots, J \right\}$$

- Project the CS on the estimated counterfactual space

$$\left[\inf_{\beta \in \tilde{\mathcal{B}}} \hat{\lambda}_\ell(\beta), \sup_{\beta \in \tilde{\mathcal{B}}} \hat{\lambda}_u(\beta) \right]$$

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- Project the CS on the estimated counterfactual space

$$\left[\inf_{\beta \in \tilde{\mathcal{B}}} \hat{\lambda}_\ell(\beta), \sup_{\beta \in \tilde{\mathcal{B}}} \hat{\lambda}_u(\beta) \right]$$

- $\bar{m}_j(\beta) = \frac{1}{n} \sum_{i=1}^n m_j(W_i; \beta)$, $\hat{\sigma}_j(\beta) = \sqrt{\frac{1}{n} \sum_{i=1}^n m_j(\cdot; \beta)^2 - \bar{m}_j^2(\beta)}$
- $\tilde{c}(\beta)$: based on full vector inference in e.g. Andrews & Soares (2010)
- $\hat{\lambda}_\ell(\beta) = \lambda_\ell^\dagger(\bar{m}_\lambda(\beta))$ for $\lambda_\ell(\beta) = \lambda_\ell^\dagger(E[m_\lambda(W; \beta)])$

Calibrated Projection

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- × Too wide: try to cover β (vector) with prob $\geq 1 - \alpha$
- × Not valid: ignore the estimation uncertainty of $\hat{\lambda}_\ell, \hat{\lambda}_u$

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- Project the CS on the estimated counterfactual space

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- × Too wide: try to cover β (vector) with prob $\geq 1 - \alpha$
- × Not valid: ignore the estimation uncertainty of $\hat{\lambda}_\ell, \hat{\lambda}_u$

My paper

- ✓ Shorter: calibrate the critical value to cover θ (scaler) with prob $\geq 1 - \alpha$
- ✓ Valid: a simple method to adjust the estimation uncertainty in $\hat{\lambda}_\ell, \hat{\lambda}_u$

Calibrated Projection

Current practice

- Get $1 - \alpha$ confidence set of β

$$\tilde{\mathcal{B}} = \left\{ \beta \in \overline{\mathcal{B}} : \frac{\sqrt{n}\tilde{m}_j(\beta)}{\hat{\sigma}_j(\beta)} \leq \tilde{c}(\beta), \quad \text{for all } j = 1, \dots, J \right\}$$

- Project the CS on the estimated counterfactual space

$$\left[\inf_{\beta \in \tilde{\mathcal{B}}} \hat{\lambda}_\ell(\beta), \sup_{\beta \in \tilde{\mathcal{B}}} \hat{\lambda}_u(\beta) \right]$$

My paper

- Get a calibrated estimator $\hat{\mathcal{B}}$

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- The CI has form

$$\left[\inf_{\beta \in \hat{\mathcal{B}}} \hat{\lambda}_\ell(\beta) - \frac{\hat{\sigma}_\ell(\beta)}{\sqrt{n}} \hat{c}(\beta), \sup_{\beta \in \hat{\mathcal{B}}} \hat{\lambda}_u(\beta) + \frac{\hat{\sigma}_u(\beta)}{\sqrt{n}} \hat{c}(\beta) \right]$$

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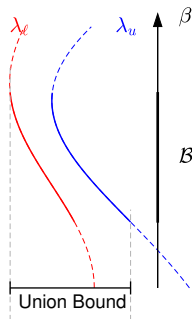
$$\left[\inf_{\beta \in \hat{\mathcal{B}}} \hat{\lambda}_\ell(\beta) - \frac{\hat{\sigma}_\ell(\beta)}{\sqrt{n}} \hat{c}(\beta), \sup_{\beta \in \hat{\mathcal{B}}} \hat{\lambda}_u(\beta) + \frac{\hat{\sigma}_u(\beta)}{\sqrt{n}} \hat{c}(\beta) \right]$$

Calibrated Critical Value: Main Idea

We need to calculate $\hat{c}(\beta)$

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Main idea: consider a small set of bounds at one time



Calibrated Critical Value: Main Idea

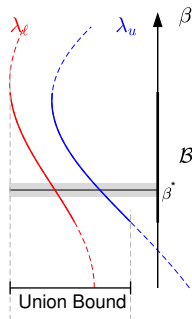
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Main idea: consider a small set of bounds at one time

1. Fix a $\beta^* \in \mathcal{B}$ and consider a local neighborhood of β^*

$$CI^{\text{local}} = \left[\inf_{\beta \in \hat{\mathcal{B}} \cap \mathcal{N}^*} \hat{\lambda}_\ell(\beta) - \frac{\hat{\sigma}_\ell(\beta)}{\sqrt{n}} \hat{c}(\beta), \sup_{\beta \in \hat{\mathcal{B}} \cap \mathcal{N}^*} \hat{\lambda}_u(\beta) + \frac{\hat{\sigma}_u(\beta)}{\sqrt{n}} \hat{c}(\beta) \right]$$



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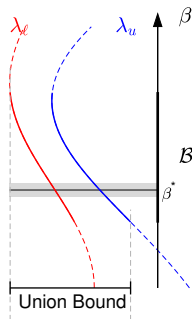
Main idea: consider a small set of bounds at one time

1. Fix a $\beta^* \in \mathcal{B}$ and consider a local neighborhood of β^*

$$CI^{\text{local}} = \left[\inf_{\beta \in \hat{\mathcal{B}} \cap \mathcal{N}^*} \hat{\lambda}_\ell(\beta) - \frac{\hat{\sigma}_\ell(\beta)}{\sqrt{n}} \hat{c}(\beta), \sup_{\beta \in \hat{\mathcal{B}} \cap \mathcal{N}^*} \hat{\lambda}_u(\beta) + \frac{\hat{\sigma}_u(\beta)}{\sqrt{n}} \hat{c}(\beta) \right]$$

2. Simplify CI^{local} by linear approximation
3. Calculate $\hat{c}(\beta^*)$ so that for $\theta \in [\lambda_\ell(\beta^*), \lambda_u(\beta^*)]$

$$P(\theta \in CI) \geq P(\theta \in CI^{\text{local}}) \geq 1 - \alpha$$



Calibrated Critical Value: Step 1 Local Neighborhood

Fix $\beta^* \in \mathcal{B}$, and define

$$CI^{\text{local}} = \left[\inf_{\beta \in \bar{\mathcal{B}} \cap \mathcal{N}^*} \hat{\lambda}_\ell(\beta) - \frac{\hat{\sigma}_\ell(\beta)}{\sqrt{n}} \hat{c}(\beta), \sup_{\beta \in \bar{\mathcal{B}} \cap \mathcal{N}^*} \hat{\lambda}_u(\beta) + \frac{\hat{\sigma}_u(\beta)}{\sqrt{n}} \hat{c}(\beta) \right]$$

where $\mathcal{N}^* = \{\beta \in \bar{\mathcal{B}} : |\beta - \beta^*| \leq r/\sqrt{n}\}$

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Rearrange terms: $\theta \in CI^{\text{local}}$ is equivalent to

$$\begin{aligned} \inf_{\beta \in \bar{\mathcal{B}} \cap \mathcal{N}^*} \frac{\hat{\lambda}_\ell(\beta) - \theta}{\hat{\sigma}_\ell(\beta)/\sqrt{n}} \leq c(\beta), \quad \text{s.t.} \quad \frac{\sqrt{n}\bar{m}_j(\beta)}{\hat{\sigma}_j(\beta)} \leq c(\beta), \quad \text{for all } j & \quad (\text{lower}) \\ \inf_{\beta \in \bar{\mathcal{B}} \cap \mathcal{N}^*} \underbrace{\frac{\theta - \hat{\lambda}_u(\beta)}{\hat{\sigma}_u(\beta)/\sqrt{n}}}_{\text{obj restriction}} \leq c(\beta), \quad \text{s.t.} \quad \underbrace{\frac{\sqrt{n}\bar{m}_j(\beta)}{\hat{\sigma}_j(\beta)}}_{\text{moment restriction}} \leq c(\beta), \quad \text{for all } j & \quad (\text{upper}) \end{aligned}$$

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And $c(\beta) \approx c(\beta^*)$

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Calibrated Critical Value: Step 2 Linearization

$\theta \in CI^{\text{local}}$ is equivalent to

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❶ For the blue part

$$\frac{\sqrt{n}\bar{m}_j(\beta)}{\hat{\sigma}_j(\beta)} \approx \mathbb{Z}_{nj}(\beta) + D_j(\beta)\sqrt{n}(\beta - \beta^*) + \frac{\sqrt{n}E[m_j(\beta^*)]}{\sigma_{Pj}(\beta^*)}$$

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$\theta \in C^{\text{local}}$ is equivalent to

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First term approx. $\frac{\sqrt{n}\bar{m}_j(\beta)}{\hat{\sigma}_j(\beta)} - \frac{\sqrt{n}E[m_j(\beta)]}{\sigma_j(\beta)}$, asymptotically normal

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$$\frac{\sqrt{n}\bar{m}_j(\beta)}{\hat{\sigma}_j(\beta)} \approx \mathbb{Z}_{nj}(\beta) + \boxed{D_j(\beta)\sqrt{n}(\beta - \beta^*)} + \frac{\sqrt{n}E[m_j(\beta^*)]}{\sigma_{Pj}(\beta^*)}$$

First term approx. $\frac{\sqrt{n}\bar{m}_j(\beta)}{\hat{\sigma}_j(\beta)} - \frac{\sqrt{n}E[m_j(\beta)]}{\sigma_j(\beta)}$, asymptotically normal

Second term approx. $\frac{\sqrt{n}E[m_j(\beta)]}{\sigma_j(\beta)} - \frac{\sqrt{n}E[m_j(\beta^*)]}{\sigma_j(\beta^*)}$, Jacobian D estimable

Calibrated Critical Value: Step 2 Linearization

$\theta \in CI^{\text{local}}$ is equivalent to

$$\inf_{\beta \in \mathcal{B} \cap \mathcal{N}^*} \frac{\hat{\lambda}_\ell(\beta) - \theta}{\hat{\sigma}_\ell(\beta)/\sqrt{n}} \leq c(\beta^*), \quad \text{s.t.} \quad \frac{\sqrt{n}\tilde{m}_j(\beta)}{\hat{\sigma}_j(\beta)} \leq c(\beta^*), \quad \text{for all } j \quad (\text{lower})$$

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$$\frac{\sqrt{n}\tilde{m}_j(\beta)}{\hat{\sigma}_j(\beta)} \approx \mathbb{Z}_{nj}(\beta) + D_j(\beta)\sqrt{n}(\beta - \beta^*) + \boxed{\frac{\sqrt{n}E[m_j(\beta^*)]}{\sigma_{Pj}(\beta^*)}}$$

Third term not consistently estimable

Use a conservative approx. $\left[\frac{\sqrt{n}\tilde{m}_j(\beta^*)}{\sqrt{\ln(n)}\hat{\sigma}_j(\beta^*)} \right]_-$, see Andrews & Soares (2010)

Calibrated Critical Value: Step 2 Linearization

$\theta \in C^{\text{local}}$ is equivalent to

$$\inf_{\beta \in \mathcal{B} \cap \mathcal{N}^*} \frac{\hat{\lambda}_\ell(\beta) - \lambda_\ell(\beta^*)}{\hat{\sigma}_\ell(\beta)/\sqrt{n}} + \frac{\lambda_\ell(\beta^*) - \theta}{\hat{\sigma}_\ell(\beta)/\sqrt{n}} \leq c(\beta^*), \quad \text{s.t.} \quad \frac{\sqrt{n}\tilde{m}_j(\beta)}{\hat{\sigma}_j(\beta)} \leq c(\beta^*), \quad \text{for all } j \quad (\text{lower})$$

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Calibrated Critical Value: Step 2 Linearization

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❷ For the red part

Calibrated Critical Value: Step 2 Linearization

$\theta \in C^{\text{local}}$ is equivalent to

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❷ For the red part

- If $\hat{\lambda}_\ell(\beta) \ll \hat{\lambda}_u(\beta)$, assume $\frac{\lambda_\ell(\beta) - \theta}{\hat{\sigma}_u(\beta)/\sqrt{n}}$ or $\frac{\theta - \lambda_u(\beta)}{\hat{\sigma}_u(\beta)/\sqrt{n}}$ is $-\infty$

Calibrated Critical Value: Step 2 Linearization

$\theta \in C^{\text{local}}$ is equivalent to

$$\inf_{\beta \in \mathcal{B} \cap \mathcal{N}^*} \underbrace{\frac{\hat{\lambda}_\ell(\beta) - \lambda_\ell(\beta^*)}{\hat{\sigma}_\ell(\beta)/\sqrt{n}} + \frac{\lambda_\ell(\beta^*) - \theta}{\hat{\sigma}_\ell(\beta)/\sqrt{n}}}_{\text{obj restriction}} \leq c(\beta^*), \quad \text{s.t.} \quad \underbrace{\frac{\sqrt{n}\bar{m}_j(\beta)}{\hat{\sigma}_j(\beta)}}_{\text{moment restriction}} \leq c(\beta^*), \quad \text{for all } j \quad (\text{lower})$$

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❷ For the red part

- If $\hat{\lambda}_\ell(\beta) \ll \hat{\lambda}_u(\beta)$, assume $\frac{\lambda_\ell(\beta) - \theta}{\hat{\sigma}_u(\beta)/\sqrt{n}}$ or $\frac{\theta - \lambda_u(\beta)}{\hat{\sigma}_u(\beta)/\sqrt{n}}$ is $-\infty$
- Otherwise, assume $\lambda_\ell(\beta) = \theta = \lambda_u(\beta)$

Calibrated Critical Value: Step 3 Calculate CV

Define

$$c(\beta^*) = \inf \left\{ c \geq 0 : P \left(\theta \in CI^{\text{local}} \right) \geq 1 - \alpha \right\}$$

Calibrated Critical Value: Step 3 Calculate CV

Define

$$c(\beta^*) = \inf \left\{ c \geq 0 : P \left(\theta \in CI^{\text{local}} \right) \geq 1 - \alpha \right\}$$

Specifically, $c(\beta^*) = \max \{ c_\ell(\beta^*), c_u(\beta^*) \}$ where

$$c_\ell(\beta^*) = \inf \{ c \geq 0 : P((\ell) \text{ and } (u) \text{ holds}) \geq 1 - \alpha \}$$

- (ℓ) and (u) are linearized version of $\theta \in CI^{\text{local}}$

$$\min_{\Delta \in \Delta(c)} \mathbb{Z}_{n,\ell}^*(\beta) + \hat{D}_{n,\ell}(\beta)\Delta \leq c \quad (\ell)$$

$$\min_{\Delta \in \Delta(c)} \mathbb{Z}_{n,u}^*(\beta) + \hat{D}_{n,u}(\beta)\Delta + \hat{\xi}_u(\beta) \leq c \quad (u)$$

where $\Delta(c)$ is the linearized moment restriction

$$\Delta(c) = \left\{ \Delta \in \sqrt{n}(\mathcal{B} - \beta) \cap r[-1, 1]^d : \mathbb{Z}^*(\beta) + \hat{D}_{n,j}(\beta)\Delta + \hat{\xi}_{n,j}(\beta) \leq c, \forall j \right\}$$

Calibrated Critical Value: Step 3 Calculate CV

Define

$$c(\beta^*) = \inf \left\{ c \geq 0 : P \left(\theta \in CI^{\text{local}} \right) \geq 1 - \alpha \right\}$$

Specifically, $c(\beta^*) = \max \{ c_\ell(\beta^*), c_u(\beta^*) \}$ where

$$c_\ell(\beta^*) = \inf \{ c \geq 0 : P((\ell) \text{ and } (u) \text{ holds}) \geq 1 - \alpha \}$$

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Solving (ℓ) and (u) is simply linear program

- Input: $\bar{m}(\beta)$, $\hat{\Sigma}(\beta)$, $\hat{D}_n(\beta)$

Calibrated Critical Value: Summary

We need to calculate $\hat{c}(\beta)$ in

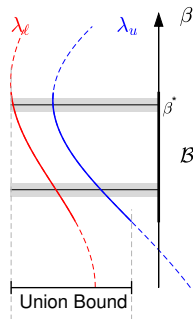
$$CI = \left[\inf_{\beta \in \hat{\mathcal{B}}} \hat{\lambda}_\ell(\beta) - \frac{\hat{\sigma}_\ell(\beta)}{\sqrt{n}} \hat{c}(\beta), \sup_{\beta \in \hat{\mathcal{B}}} \hat{\lambda}_u(\beta) + \frac{\hat{\sigma}_u(\beta)}{\sqrt{n}} \hat{c}(\beta) \right]$$

Main idea: consider a small set of bounds at one time

1. Fix a $\beta^* \in \mathcal{B}$ and consider CI^{local}
2. Simplify CI^{local} by linear approximation
3. Calculate $\hat{c}(\beta^*)$ so that for $\theta \in [\lambda_\ell(\beta^*), \lambda_u(\beta^*)]$

$$P(\theta \in CI) \geq P(\theta \in CI^{\text{local}}) \geq 1 - \alpha$$

- Solving linear programs
4. Fix another β^* and repeat $\Rightarrow$ get $\hat{c}(\beta)$ for all β
 - E-A-M algorithm in Kaido, Molinari, Stoye (2019) (modified)



Asymptotic Size Properties

Assumption

- ① *Lipschitz Continuous (LC)*. $\|D_P(\beta) - D_P(\tilde{\beta})\| \leq M\|\beta - \tilde{\beta}\|$, $\|D_P(\beta)\| \leq M$
- ② *Consistent Estimator (CE)*. $\hat{D}(\cdot) \xrightarrow{P} D_P(\cdot)$, $\hat{\Sigma}(\cdot) \xrightarrow{P} \Sigma_P(\cdot)$
- ③ *Rank Condition (RC)*. $\inf_{\beta} \text{eig}(\Sigma(\beta)) \geq \omega > 0$

Theorem (Uniform Coverage)

Suppose Assumptions LC, CE, RC and regularity conditions hold. Let $0 < \alpha < 1/2$. Then

$$\liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}} \inf_{\theta \in \bigcup_{\beta \in B} [\lambda_{P,\ell}(\beta), \lambda_{P,u}(\beta)]} P(\theta \in CI) \geq 1 - \alpha.$$

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This theorem holds under weaker conditions

- Plug in estimators ► Plug-in
- Covariance matrix has singular limit

Outline

- 1 Preview of the Contributions
- 2 Inference with Finite $\mathcal{B}$ (e.g. Difference in Differences)
- 3 Inference with Infinite $\mathcal{B}$ (e.g. Structural Counterfactual)
- 4 Conclusion

Conclusion

This is the first paper that systematically studies inference on **union bounds**

$$\bigcup_{\beta \in \mathcal{B}} [\lambda_\ell(\beta), \lambda_u(\beta)]$$

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Finite $\mathcal{B}$: e.g. DiD

I propose a CI based on *modified conditional* inference

Valid and **Short**:

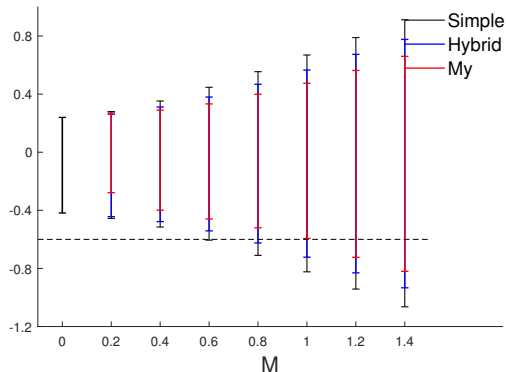
- Theory & Simulation: shorter CI & larger local power under a large set of DGPs

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Valid and **Short**:

- Computationally feasible: solving linear programs
- Wide applications on structural counterfactuals

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Infinite $\mathcal{B}$: e.g. structural counterfactual

	Valid	Calibrated	All Firms	Large Firms	Small Firms
DM18	×	×	[-10.2, -6.1]	[-17.3, -12.7]	[0.3, 0.5]
DM18 adj	✓	×	[-22.9, 1.5]	[-30.4, -0.8]	[-0.1, 1.4]
New	✓	✓	[-20.4, -1.6]	[-27.4, -8.7]	[0, 1.2]

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► RDD

► Bunching

Finite $\mathcal{B}$: Empirical Illustration

To study the employment and wage effect, run

$$\begin{aligned}\log(\text{emp}_{rt}) &= \sum_{\tau=2011, \tau \neq 2014}^{2016} \gamma_{\tau}^e \overline{GAP}_r 1[\tau = t] + \alpha_r^e + \zeta_t^e + \varepsilon_{rt}^e \\ \log(\text{wage}_{rt}) &= \sum_{\tau=2011, \tau \neq 2014}^{2016} \gamma_{\tau}^w \overline{GAP}_r 1[\tau = t] + \alpha_r^w + \zeta_t^w + \varepsilon_{rt}^w\end{aligned}$$

- $\log(\text{emp}_{rt})$ is the log employment in district r time t ; $\log(\text{wage}_{rt})$ is log wage
- $\overline{GAP}_r$ is a measure of the exposure to the minimum wage
- Pre-policy year 2011-2014

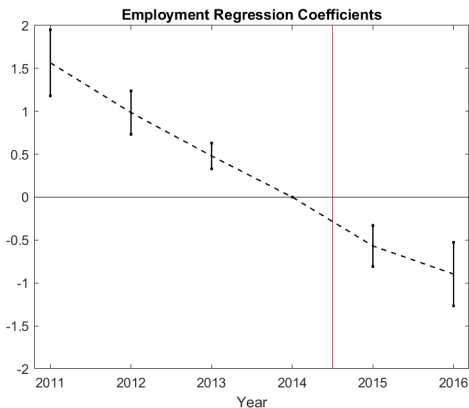
We are interested in the employment and wage effect at $t = 2015$ [▶ Back](#)

Finite $\mathcal{B}$: Empirical Illustration

RR23 in Dustmann, Lindner, Schönberg, Umkehrer, Vom Berge (2022, QJE)

What are the effects of the minimum wage? [▶ Back](#)

- Is employment effect ≥ -0.6 (wage effect) without parallel trends?



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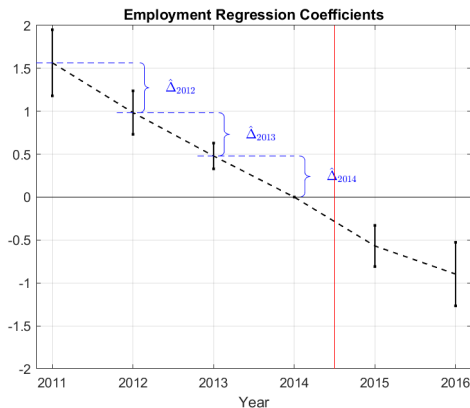


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Relax the parallel trends assumption: $|\Delta_{2015} - \Delta_{2014}| \leq M \cdot \max_{t=2013,2014} |\Delta_t - \Delta_{t-1}|$

Infinite $\mathcal{B}$

Dickstein and Morales (2018, QJE) [▶ Back](#)

If we change the information set from small to large, what is the percentage change in number of exporters

Infinite $\mathcal{B}$

Dickstein and Morales (2018, QJE) [▶ Back](#)

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Firm	All	Large	Small
DM18	[-10.2, -6.1]	[-17.3, -12.7]	[0.3, 0.5]

DM18 CI is calculated by

$$CI^{DM18} = \bigcup_{\beta \in \tilde{\mathcal{B}}_{1-\alpha}} \left[\hat{\lambda}_\ell(\beta), \hat{\lambda}_u(\beta) \right]$$

where $\tilde{\mathcal{B}}_{1-\alpha}$ is a valid $1 - \alpha$ CS for β

Infinite $\mathcal{B}$

Dickstein and Morales (2018, QJE) [▶ Back](#)

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where $\tilde{\mathcal{B}}_{1-\alpha}$ is a $1 - \alpha$ CS for β assuming φ is known

- The counterfactual depends on the DGP
- The model contains plug-in estimators

Infinite $\mathcal{B}$

Dickstein and Morales (2018, QJE) [▶ Back](#)

If we change the information set from small to large, what is the percentage change in number of exporters

Firm	All	Large	Small
DM18	[-10.2, -6.1]	[-17.3, -12.7]	[0.3, 0.5]
DM18 adj	[-22.9, 1.5]	[-30.4, -0.8]	[-0.1, 1.4]

DM18 adj CI follows Bonferroni correction

$$CI^{adj} = \left[\inf_{\beta \in \hat{\mathcal{B}}_{1-\alpha/2}} \hat{\lambda}_{\ell}(\beta) - \Phi^{-1}\left(1 - \frac{\alpha}{4}\right) \frac{\hat{\sigma}_{\ell}(b)}{\sqrt{n}}, \quad \sup_{\beta \in \hat{\mathcal{B}}_{1-\alpha/2}} \hat{\lambda}_u(b) + \Phi^{-1}\left(1 - \frac{\alpha}{4}\right) \frac{\hat{\sigma}_u(b)}{\sqrt{n}} \right]$$

where $\hat{\mathcal{B}}_{1-\alpha/2}$ is a valid $1 - \frac{\alpha}{2}$ CS for β

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Dickstein and Morales (2018, QJE) [▶ Back](#)

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Simulation - Setting

I set $n = 5000$

Each sample $\{W_i\}_{i=1}^n$ and estimator is generated by

$$\begin{pmatrix} W_{\Delta,i} \\ W_{\gamma,i} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \Delta \\ \gamma \end{pmatrix}, n\Omega \right)$$

The estimator is calculated by

$$\begin{pmatrix} \hat{\Delta} \\ \hat{\gamma} \end{pmatrix} = \begin{pmatrix} \frac{1}{n} \sum W_{\Delta,i} \\ \frac{1}{n} \sum W_{\gamma,i} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \Delta \\ \gamma \end{pmatrix}, \Omega \right)$$

I conduct inference using pair $(\hat{\Delta}, \hat{\gamma}, \Omega)$

► Back

Example - Dickstein and Morales (2018, QJE)

Firms $i = 1, \dots, N$ decide whether to export to a foreign market f [▶ Back](#)

- The home and foreign market revenue is determined by

$$r_h = X_1 + X_2, \quad r_f = \varphi r_h + e$$

- Let $\mathcal{J}$ be exporter's info set. The expected export profit and the decision are

$$p_f = E[r_f | \mathcal{J}] - \beta_2 - \beta_1 \nu, \quad d = 1 \left[\beta_1^{-1} E[r_f | \mathcal{J}] - \beta_1^{-1} \beta_2 \geq \nu \right]$$

- Distributional assumption $(X_1, X_2, e, \nu) \sim \mathcal{N}(0, \mathcal{I}_4)$
- The researcher observes X_1, X_2, r_h, d, dr_f

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The plug-in estimand φ

$$\varphi = \frac{\text{cov}(r_f, r_h | d = 1)}{\text{var}(r_h | d = 1)} = \frac{E[r_f r_h d] E[d] - E[r_h d] E[r_f d]}{E[r_h^2 d] E[d] - E[r_h d]^2}$$

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- The researcher observes X_1, X_2, r_h, d, dr_f

The identified set $\mathcal{B}$ for the structural parameter (β_1, β_2)

- Revealed preference moment condition $E[m(W; \beta, \varphi)] \leq 0$

$$m(W; \beta, \varphi) = (1 - d) \underbrace{\left(\beta_1^{-1} \varphi r_h - \beta_1^{-1} \beta_2 \right)}_{\text{export profit}} - d \underbrace{\frac{\phi \left(\beta_1^{-1} \varphi r_h - \beta_1^{-1} \beta_2 \right)}{\Phi \left(\beta_1^{-1} \varphi r_h - \beta_1^{-1} \beta_2 \right)}}_{\text{selection bias}}$$

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- The researcher observes X_1, X_2, r_h, d, dr_f

The counterfactual of interest θ

- Example 1: change in # exporters if $\mathcal{J}$ changes from $\{X_1\}$ to $\{X_1, X_2\}$

$$\lambda_\ell(\beta, \varphi) = \lambda_u(\beta, \varphi) = \frac{E \left[\Phi \left(\beta_1^{-1} \varphi r_h - \beta_1^{-1} \beta_2 \right) \right]}{E \left[\Phi \left(\beta_1^{-1} \varphi X_1 - \beta_1^{-1} \beta_2 \right) \right]} - 1$$

- For a given β , θ is *unknown* but *point identified*

Example - Dickstein and Morales (2018, QJE)

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- The researcher observes X_1, X_2, r_h, d, dr_f

The counterfactual of interest θ

- Example 2: change in # exporters if β_2 changes to $1.2\beta_2$

$$\lambda_\ell(\beta, \varphi) \leq \lambda_u(\beta, \varphi)$$

formed by Jensen's inequality since $\mathcal{J}$ is unobservable

- For a given β , θ is *unknown* and *partially identified*

Bunching and Taxable Income Elasticity

Blomquist, Newey, Kumar and Liang (2021, JPE)

▶ Intro

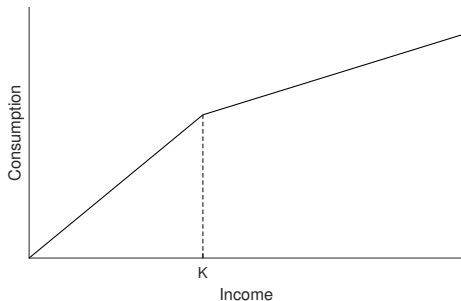
▶ Conclusion

Bunching and Taxable Income Elasticity

Blomquist, Newey, Kumar and Liang (2021, JPE)

▶ Intro

▶ Conclusion



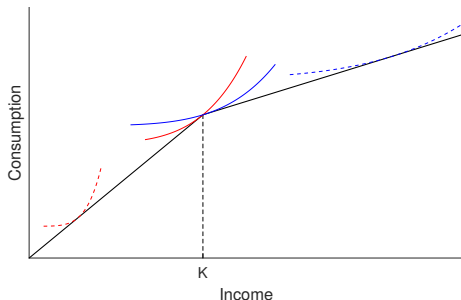
The consumption has two linear segments with slopes $\rho_1 > \rho_2$ and a kink at K

Bunching and Taxable Income Elasticity

Blomquist, Newey, Kumar and Liang (2021, JPE)

Intro

Conclusion



Assume isoelastic utility function $U(c, y, \eta) = c - \frac{\eta}{1+1/\theta} \left(\frac{y}{\eta}\right)^{1+1/\theta}$

- $\eta > 0$: unobservable heterogeneity
- $\theta > 0$: income elasticity
- $\phi(\eta)$: be the density of η

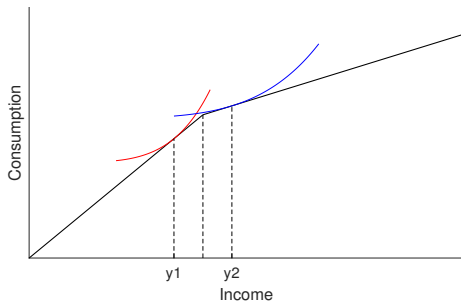
No restriction on $\phi(\eta) \Rightarrow \theta$ is not identified

Bunching and Taxable Income Elasticity

Blomquist, Newey, Kumar and Liang (2021, JPE)

► Intro

► Conclusion



Assume there are $\sigma_2 \geq 1 \geq \sigma_1 > 0$ such that

$$\sigma_1 \min \{ \phi(\eta_1), \phi(\eta_2) \} \leq \phi(\eta) \leq \sigma_2 \max \{ \phi(\eta_1), \phi(\eta_2) \}$$

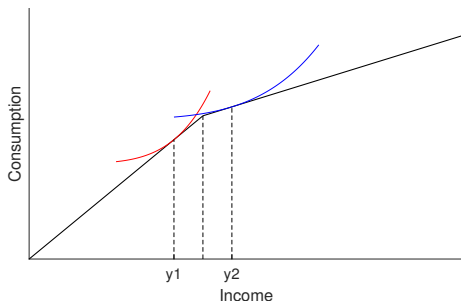
for $\eta \in [\eta_1, \eta_2]$

Bunching and Taxable Income Elasticity

Blomquist, Newey, Kumar and Liang (2021, JPE)

Intro

Conclusion



The identified set is $\theta \in [\min_{b=1,2} \lambda_{l,b}, \max_{b=1,2} \lambda_{u,b}]$ where

$$\lambda_{l,1} = \frac{\log \left(\frac{y_1}{y_2} + \frac{P(y_1 \leq Y \leq y_2)}{f^-(y_1)\sigma_2 y_2} \right)}{\log \rho_1 - \log \rho_2}, \quad \lambda_{l,2} = \frac{-\log \left(\frac{y_2}{y_1} - \frac{P(y_1 \leq Y \leq y_2)}{f^-(y_2)\sigma_2 y_1} \right)}{\log \rho_1 - \log \rho_2}$$

Regression Discontinuity Design

Kolesár and Rothe (2018, AER)

▸ Intro

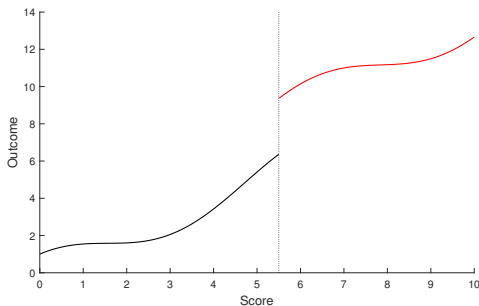
▸ Conclusion

Regression Discontinuity Design

Kolesár and Rothe (2018, AER)

▶ Intro

▶ Conclusion



Treatment $D = 1 \{X \geq k\}$ with running variable X

Let $\mu(X) = E[Y | X]$. The ATE at the threshold is

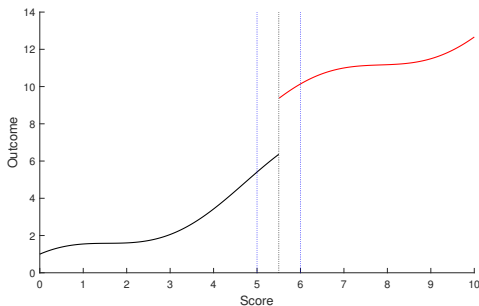
$$\theta = E[Y(1) - Y(0) | X = 0] = \lim_{x \downarrow k} \mu(x) - \lim_{x \uparrow k} \mu(x)$$

Regression Discontinuity Design

Kolesár and Rothe (2018, AER)

▶ Intro

▶ Conclusion



Estimate θ : local OLS of Y on $m(X)$ with $X \in [k - h, k + h]$ where

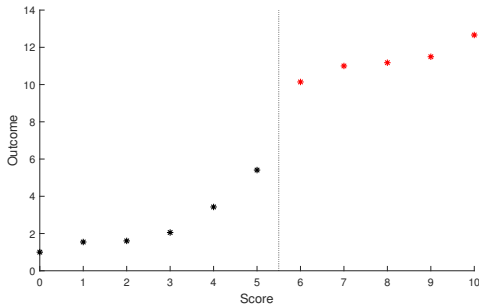
$$m(x) = (1\{x \geq k\}, 1\{x \geq k\}(x - k), \dots, 1\{x \geq k\}(x - k)^p, \\ 1, x - k, \dots, (x - k)^p)$$

Regression Discontinuity Design

Kolesár and Rothe (2018, AER)

► Intro

► Conclusion

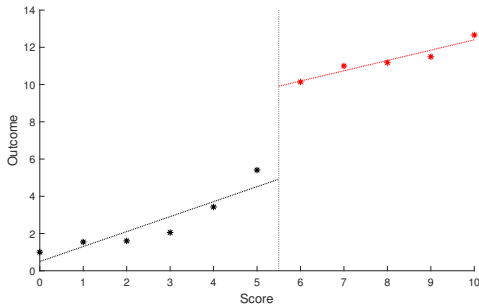


Regression Discontinuity Design

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▶ Intro

▶ Conclusion

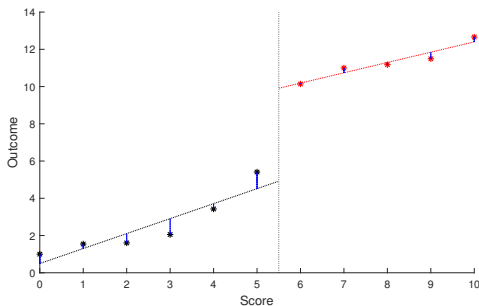


Regression Discontinuity Design

Kolesár and Rothe (2018, AER)

► Intro

► Conclusion



Assumption: bounds on specification errors at the threshold

$$|\lim_{x \uparrow k} \Delta(x)| \leq \max_{x' < k} |\Delta(x')|, \quad |\lim_{x \downarrow k} \Delta(x)| \leq \max_{x' > k} |\Delta(x')|$$

Size & Power: Comparison with Simple CI

Theorem (Symmetric Or Large Bounds)

Suppose Assumptions KS, AN, FR, CE hold, $\alpha \in (0, \frac{1}{2})$

(Symmetric Bounds) If $\text{corr}(\hat{\lambda}_{\ell, b_1}, \hat{\lambda}_{\ell, b_2}) < \rho_1^*(\alpha, \alpha^c)$, $\text{corr}(\hat{\lambda}_{\ell, b_\ell}, \hat{\lambda}_{\ell, b_u}) < \rho_2^*(\alpha)$

$$\hat{\lambda}_\ell = \hat{\lambda}_u$$

❶ My CI is Strictly Shorter:

There is $\alpha' > \alpha$ such that

$$\liminf_n \inf_{P \in \mathcal{P}_n} P \left(CI^m(\hat{\lambda}_n, \hat{\Sigma}_n/n; \alpha) \subseteq CI^{sim}(\hat{\lambda}_n, \hat{\Sigma}_n/n; \alpha') \right) = 1$$

❷ My CI has Higher Power:

For all $P_n \in \mathcal{P}_n$, there is a subsequence P_{τ_n} and $\kappa \in (0, +\infty)$ such that

$$\liminf_n P_{\tau_n}(\theta_{\tau_n} \notin CI^m(\hat{\lambda}, \hat{\Sigma}/\tau_n; \alpha)) - P_{\tau_n}(\theta_{\tau_n} \notin CI^{sim}(\hat{\lambda}, \hat{\Sigma}/\tau_n; \alpha)) > 0$$

for some $\theta_{\tau_n} = \theta_\ell - \frac{\kappa}{\sqrt{\tau_n}}$.

Size & Power: Comparison with Simple CI

Theorem (Symmetric Or Large Bounds)

Suppose Assumptions KS, AN, FR, CE hold, $\alpha \in (0, \frac{1}{2})$

(Large Bounds) Let $\kappa_n = o(\sqrt{n})$ and $\kappa_n \rightarrow \infty$, and

$$\mathcal{P}_n = \left\{ P \in \mathcal{P} : \lambda_{u,b_u} - \lambda_{\ell,b_\ell} \geq \frac{\kappa_n}{\sqrt{n}} \right\}$$

❶ *My CI is Strictly Shorter:*

There is $\alpha' > \alpha$ such that

$$\liminf_n \inf_{P \in \mathcal{P}_n} P \left(CI^m(\hat{\lambda}_n, \hat{\Sigma}_n/n; \alpha) \subseteq CI^{sim}(\hat{\lambda}_n, \hat{\Sigma}_n/n; \alpha') \right) = 1$$

❷ *My CI has Higher Power:*

For all $P_n \in \mathcal{P}_n$, there is a subsequence P_{τ_n} and $\kappa \in (0, +\infty)$ such that

$$\liminf_n P_{\tau_n}(\theta_{\tau_n} \notin CI^m(\hat{\lambda}, \hat{\Sigma}/\tau_n; \alpha)) - P_{\tau_n}(\theta_{\tau_n} \notin CI^{sim}(\hat{\lambda}, \hat{\Sigma}/\tau_n; \alpha)) > 0$$

for some $\theta_{\tau_n} = \theta_\ell - \frac{\kappa}{\sqrt{\tau_n}}$.

E-A-M Algorithm

Main idea:

- Fit the m

► Back

Plug-in Estimators

The moment inequalities are

$$E[m_j(W; \beta, \varphi)] \leq 0, \quad j = 1, \dots, J, \quad \varphi = \varphi^\dagger(E[m_\varphi(W)])$$

Decompose the variance: $\sqrt{n}(\bar{m}(\beta, \hat{\varphi}) - E[m(W; \beta, \varphi)])$ is approximately

1. $\sqrt{n}(\bar{m}(\beta, \varphi) - E[m(W; \beta, \varphi)])$
2. $\sqrt{n}(E[m(W; \beta, \hat{\varphi})] - E[m(W; \beta, \varphi)])$

Let

$$G(\beta) = \begin{bmatrix} \mathcal{I}_J & \nabla_{\varphi'} E[m(X; \beta, \varphi)] \nabla_{m'_\varphi} \varphi^\dagger(m_\varphi) \end{bmatrix}$$
$$\Omega(\beta) = \text{var} \begin{pmatrix} m'(X; \beta, \varphi), & m'_\varphi(X) \end{pmatrix}$$

The asymptotic variance of $\sqrt{n}(\bar{m}(\beta, \hat{\varphi}) - E[m(W; \beta, \varphi)])$ is

$$\Sigma_P(\beta) = G_P(\beta) \Omega_P(\beta) G'_P(\beta)$$

Size & Power: Assumptions

Assumption (Known Singularity (KS))

There are known $|\mathcal{B}| \times J$ matrices A_ℓ, A_u such that for some $(\delta_P, \hat{\delta}_n)$

$$\lambda_\ell = A_\ell \delta_P, \lambda_u = A_u \delta_P, \hat{\lambda}_\ell = A_\ell \hat{\delta}_n, \hat{\lambda}_u = A_u \hat{\delta}_n$$

Assumption (Asymptotic Normality (AN))

Let $\xi_P \sim \mathcal{N}(0, \Omega_P)$. Assume

$$\lim_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} \sup_{f \in BL_1} \left| E_P \left[f \left(\sqrt{n} \left(\hat{\delta}_n - \delta_P \right) \right) \right] - E \left[f(\xi_P) \right] \right| = 0$$

Assumption (Full Rank (FR))

For all $P \in \mathcal{P}$, $0 < \underline{e} \leq \text{eig}(\Omega_P) \leq \bar{e} < \infty$.

Assumption (Consistent Covariance Estimator (CE))

For all $\varepsilon > 0$, $\lim_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} P \left(\left\| \hat{\Omega}_n - \Omega_P \right\| > \varepsilon \right) = 0$